NovaTech Industries — Checkpoint 2.8

Supplier Performance Metrics & Category Compliance

Status: Implementation exercise | **Data version:** v2 | **Reporting date:** 2026-10-21

Learning objectives

• Aggregate historical transactions into supplier-level quality and reliability indicators.

• Use a fixed 12-month reporting window and explicit denominators.

• Flag insufficient evidence instead of inventing scores.

• Keep category-specific compliance separate from supplier-wide performance.

• Validate input data and produce auditable CSV outputs.

Confirmed source schemas

• `suppliers.csv`: `supplier_id, supplier_name, country, status` (12 rows)

• `products.csv`: `product_id, sku, product_name, category, unit_of_measure` (20 rows)

• `supplier_products.csv`: `supplier_id, product_id, minimum_order_quantity, available_quantity` (45 rows)

• `supplier_deliveries.csv`: `delivery_id, supplier_id, planned_delivery_date, actual_delivery_date, on_time, status` (177 rows)

• `supplier_quality_inspections.csv`: `inspection_id, supplier_id, inspection_date, inspected_units, accepted_units, rejected_units` (66 rows)

• `supplier_compliance.csv`: `supplier_id, category, approval_status, valid_until` (26 rows)

Metric definitions

Only events dated between **2025-10-21 and 2026-10-21**, inclusive, count. Completed deliveries use **actual delivery date**; inspections use **inspection date**.

• **Quality (%)** = 100 × accepted units / inspected units.

• **Reliability (%)** = 100 × on-time completed deliveries / completed deliveries.

• **On time** = actual delivery date on or before planned delivery date.

• **Proposed evidence policy:** at least 100 inspected units AND at least 10 completed deliveries. These are *proposed, configurable thresholds*, not yet formally approved.

• **Evidence status:** `EVIDENCE_SUFFICIENT` or `REVIEW_REQUIRED`. Missing-evidence scores are blank rather than estimated. Each score is independently withheld when its own evidence threshold is unmet.

• **Compliance:** evaluate the supplier for each product category it actually supplies. `VALID` means an approved/active compliance record with an unexpired `valid_until`; otherwise `REVIEW_REQUIRED` or `MISSING`. A compliance status is **not** a purchasing authorization.

Save and execute

1. Save `analyze_supplier_performance_v2.py` to `src/analysis/analyze_supplier_performance_v2.py`.

2. From the project root, run:

```
mkdir -p src/analysis
python3 src/analysis/analyze_supplier_performance_v2.py
```

The script reads from `data/synthetic/v2/` and writes to `data/analysis/v2/`:

• `supplier_performance_metrics.csv`: one row per supplier (expected **12**).

• `supplier_category_compliance.csv`: one row per supplier-category combination, count dependent on actual relationships.

The input datasets are never overwritten. The generator is deterministic for a fixed input and reporting date.

Validation exercise

```
python3 - <<'PY'
import csv
from pathlib import Path
base = Path('data/analysis/v2')
for filename in ['supplier_performance_metrics.csv', 'supplier_category_compliance.csv']:
    with (base / filename).open(newline='', encoding='utf-8') as f:
        rows = list(csv.DictReader(f))
    print(filename, 'rows:', len(rows))
    print('first record:', rows[0] if rows else '(empty)')
PY
```

Check `SUP012` for low-history behavior. Confirm that scores are blank when evidence is insufficient. Inspect the category compliance output: missing, expired, or pending approval must not be labeled valid.

Important design limits

• **No full quotation eligibility or weighted ranking yet.** This checkpoint creates evidence used by a later integration step.

• Status values are normalized case-insensitively. Completed deliveries are identified by `completed` or `delivered`; other statuses are excluded.

• The script checks that reported on-time flags agree with actual/planned dates; inconsistent synthetic records raise an explicit error for investigation.

• The category compliance logic recognizes `approved` or `active` as affirmative approval values. Verify this assumption against the generator's actual statuses before relying on results.

• A supplier may have adequate history yet still be inactive or noncompliant. Those checks must remain separate.

Next checkpoint

2.9 — Integrate supplier performance, category compliance, and quotation eligibility, then implement cost and lead-time normalization and the 40/25/20/15 scoring formula only for fully eligible offers.